MOHSIN KHAN

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Professional Summary

Final-year B.Tech student in Artificial Intelligence & Data Science with hands-on experience in machine learning, software development, data analysis, and AI-powered applications. Strong foundation in Python, C++, SQL, model evaluation, and building practical end-to-end systems.

Education

Madhav Institute of Technology & Science (MITS), Gwalior — *Expected May 2027*
B.Tech, Artificial Intelligence & Data Science — Final-year student

Jawahar Navodaya Vidyalaya, Madhya Pradesh — *2022*
Class XII (PCM)

Technical Skills

Languages	Python, C++, SQL, Bash
AI & Machine Learning	PyTorch, scikit-learn, Hugging Face Transformers, NLP, spaCy, OpenCV, MediaPipe
Generative AI & LLMs	LangChain, LangGraph, LlamaIndex, LiteLLM, Ollama, RAG, FAISS, ChromaDB, LoRA/PEFT
Data & Tools	NumPy, Pandas, Matplotlib, Seaborn, PostgreSQL, MongoDB, SQLite, FastAPI, pytest, Docker, Git, Linux

Selected Projects

NeuroFroge — Neural Network Framework from Scratch

C++ · Linear Algebra · Deep Learning · Performance Optimization

- Built a C++ neural-network framework with N-dimensional tensors, broadcasting, indexing, matrix multiplication, and stride-based zero-copy reshape/transpose views.
- Implemented linear layers, ReLU, Sigmoid, Tanh, Softmax, cross-entropy, backpropagation, gradient accumulation, SGD, and mini-batch training.
- Trained a 784→128→10 Fashion-MNIST classifier at approximately 81% test accuracy; CPU optimizations achieved about 1.76 seconds for a five-epoch experiment.

Transformer Fine-Tuning: BERT & Dolly 15K

Python · PyTorch · Hugging Face Transformers · PEFT · LoRA

- Fine-tuned BERT on IMDB reviews for binary sentiment classification, achieving ROC-AUC 0.9675, precision 92.57%, and recall 94.64%.
- Fine-tuned Qwen2.5-0.5B-Instruct on Databricks Dolly 15K for instruction tasks including Q&A, summarization, classification, and information extraction.
- Applied LoRA (rank 8, alpha 16) with PEFT, training approximately 540K of 494.6M parameters (0.11%); further experiments: LLM Fine-Tuning Experiments.

Local AI Agent — Modular Personal AI Assistant

Python · LangGraph · LangChain · Ollama · Groq · LlamaIndex · Playwright · SQLite

- Developed hybrid local/cloud LLM routing with privacy-aware local inference, automatic fallback, and a circuit breaker for repeated cloud failures.
- Integrated RAG, email, calendar, tasks, web search, file discovery, expense tracking, and Playwright browser automation with human approval.
- Added persistent user memory and CLI–Telegram coordination with file delivery; reduced the system prompt from about 570 to 206 tokens (64%).

Achievements

- Advanced to Round 2 in Smart India Hackathon, IIIT Delhi Hackathon, and IIIT Gwalior Hackathon.
- Completed the Aspire Leaders Program (2024); earned 5-star HackerRank badges in Python, SQL, and C++.

Certifications & Coursework

- **Generative AI for Computer Vision** — NPTEL, IIT Guwahati — Jul–Oct 2026
- **Deep Learning for Natural Language Processing** — NPTEL, IIT Kharagpur — Jul–Oct 2026
- **Business Intelligence & Analytics** — NPTEL — Apr 2026
- **Oracle Agentic AI Certified Foundations Associate** — Oracle — Aug 2026
- **Introduction to LangChain — Python** — LangChain Foundation — Aug 2026–Aug 2028
- **LangGraph Essentials — Python** — LangChain Foundation — Jun 2026
- **SQL (Basic) Certificate** — HackerRank — May 2026